

Frede Emnetu . Jarvis Consulting

Backend-leaning Full-Stack Developer specializing in Java/Spring Boot and React, with hands-on experience shipping production systems across e-commerce, workflow automation, and QA automation. Comfortable owning backend architecture end-to-end – from API design and database optimization to CI/CD deployment – while collaborating directly with stakeholders to define requirements and deliver working software.

Skills

Proficient: Java, Linux/Bash, RDBMS/SQL, Agile/Scrum, Git, Spring Boot, Docker, API Testing

Competent: Next.js, C++, JavaScript, C#, React

Familiar: Jenkins, AWS Cloud management, Test/Behaviour-Driven Development, Rust, MongoDB

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_FredeEmnetu

Cluster Monitor [GitHub]: Developed a Linux Cluster Monitoring Agent (LCA) intended to be used as a resource monitoring solution designed to track and record the hardware specifications and real-time usage statistics of 3+ Linux nodes within a server cluster. The system was built with bash scripts and CRON for scheduling. The system automatically collects CPU, memory, and disk data from each host and persists it to a centralized PostgreSQL database for historical analysis and reporting.

Java Grep Application [GitHub]: Built a Java command-line utility replicating Linux grep, recursively searching a directory tree and matching file lines against a user-supplied regex using java.nio and java.util.regex, with SLF4J/Log4j logging. Identified a heap-memory risk in the initial implementation and outlined a fix path via JVM heap sizing and a planned refactor to lazy stream processing. Containerized the app with Docker on an Alpine base image and published it to Docker Hub for consistent, JDK-free deployment. Verified functionality through manual testing against sample files, including edge cases like empty directories and full-match regex patterns.

Python Data Analysis [GitHub]: Delivered a proof-of-concept customer analytics project for a UK giftware retailer facing stagnant revenue, using Python, Pandas, and NumPy in a Jupyter Notebook connected to a Dockerized PostgreSQL warehouse. Performed RFM segmentation across ~5,900 customers into 10 behavioral segments, finding the top 5% of customers generated 52% of revenue, with 754 “At Risk” and 86 “Can’t Lose” customers showing churn signs. Visualized findings with Matplotlib and delivered actionable recommendations (VIP retention, win-back campaigns, seasonal timing) to the marketing team. Presented results as both a technical notebook and an executive presentation.

Java Stock Quote App [GitHub]: Built a terminal-based Java trading application integrating the Alpha Vantage API for live stock data, using JDBC to persist quotes, positions, and trading activity to a PostgreSQL database, with OkHttp for HTTP requests and Jackson for JSON parsing. Applied the DAO pattern to separate CRUD logic from business logic, keeping the service layer decoupled from SQL and testable via mocked dependencies. Validated functionality with JUnit 4 and Mockito unit tests against mocked DAOs, plus integration tests against a live Dockerized PostgreSQL instance. Packaged the app with Maven and Docker for consistent setup and deployment.

Databricks & Spark [GitHub]: Built two Medallion (Bronze → Silver → Gold) data pipelines in Databricks using PySpark and Delta Lake on Azure, turning raw data into analysis-ready datasets for dashboards. Ingested 13M+ financial transaction rows from Azure SQL Database via JDBC, then cleaned, deduplicated, and type-cast them in the Silver layer, with each stage gated on the success of the previous one, to produce Gold tables for fraud and user-behavior analysis. Developed a second pipeline using Lakeflow Declarative Pipelines (DLT) and Auto Loader to incrementally ingest daily Alpha Vantage data for four tickers, computing 5/30/90-day percent change and average volume with PySpark window functions in a Gold materialized view. Governed all tables with Unity Catalog on Azure Data Lake Storage and served results through Databricks dashboards.

Spring Boot [GitHub]: Built a Spring Boot REST API that simulates a securities trading platform, supporting trader and account management, real-time quotes from the Finnhub API, and buy/sell market orders validated against account balance and position size. Structured the app into controller, service, and DAO layers, using Spring Data JPA/Hibernate with PostgreSQL for persistence and Apache HttpClient for market data, and cached quotes locally to avoid repeated external API calls. Verified behavior with JUnit 5 and Mockito unit tests on the service layer, plus integration tests against a real PostgreSQL database and live Finnhub calls, and documented the endpoints with Swagger UI. Containerized the app and database with multi-stage Docker builds on a shared bridge network and published both images to Docker Hub.

Highlighted Projects

C++ Multithreading: Built a multithreaded C++ web scraping application using parallel processing, reducing data retrieval time by 40% while enabling scalable, concurrent data collection across multiple websites via Ubuntu Linux; Documented codebase that improved readability and enhanced maintainability and resource efficiency.

Springboot E-commerce Store: Built and deployed a production e-commerce platform using Java/Spring Boot, MongoDB, and React, supporting 1,000+ products; improved search performance by 35% through query tuning and indexing, implemented a Jenkins/Docker CI/CD pipeline with Azure deployment to boost release reliability by 30%, and collaborated with stakeholders to design secure REST APIs, data models, and authentication flows aligned with data protection standards.

Professional Experiences

Software Developer, Jarvis (2026-present): Developed a series of projects spanning multiple languages, frameworks, and technologies to build practical, hands-on software development experience. Engineered backend applications using Java and PostgreSQL, leveraging Docker for containerization and Google Cloud VMs for cloud-based deployment. Automated workflows and system tasks through Bash scripting within a Linux environment, reinforcing core DevOps principles throughout each project.

Software Development Engineer in Test, QA Consultants (2023): Built a fully automated test suite in C#, Gherkin, and SpecFlow, reaching a 93% pass rate and cutting production defects by 20%. Integrated the test suite into the existing CI/CD pipeline, accelerating release cadence without increasing manual QA load. Grew Selenium-based UI coverage, materially reducing regression surface across Agile sprints.

Education

Ontario Tech University (2018-2022), Bachelor of Science, Computer Science - Awards of Recognition Scholarship - President's List (2021): Obtained a 3.8 GPA - Dean's List (2019,2022): Maintained above a 3.5 GPA - GPA: 3.72/4.3

Miscellaneous

- Oracle Cloud Infrastructure 2025 Certified Foundations Associate (2025)
- miniature painting
- Fitness & Strength training